Line Charts: Theory & Best Practices

Workshop 6 — Module 2

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Data Visualization for Data Scientists
UNYT Tirana

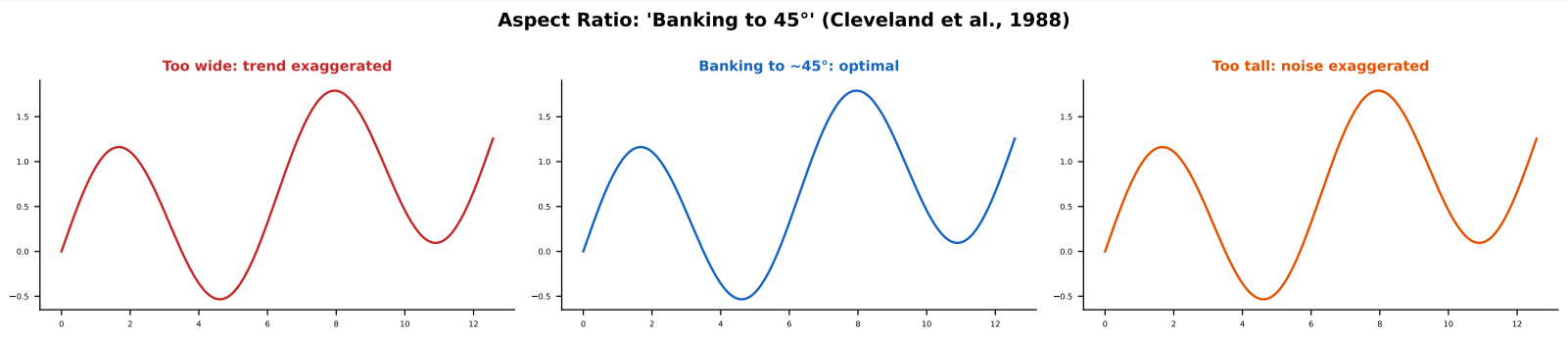
Module Roadmap

1. Aspect ratio: banking to 45° (Cleveland, 1988)
2. The dual y-axis debate: why it usually misleads
3. Indexed lines: normalising for fair comparison
4. Grey+accent strategy for multiple series
5. Five design principles for publication-quality line charts

Learning Outcome
You will apply aspect ratio principles, avoid dual y-axes, index multiple series for fair comparison, and use grey+accent to highlight the narrative in multi-line charts.

Aspect Ratio

Banking to 45°

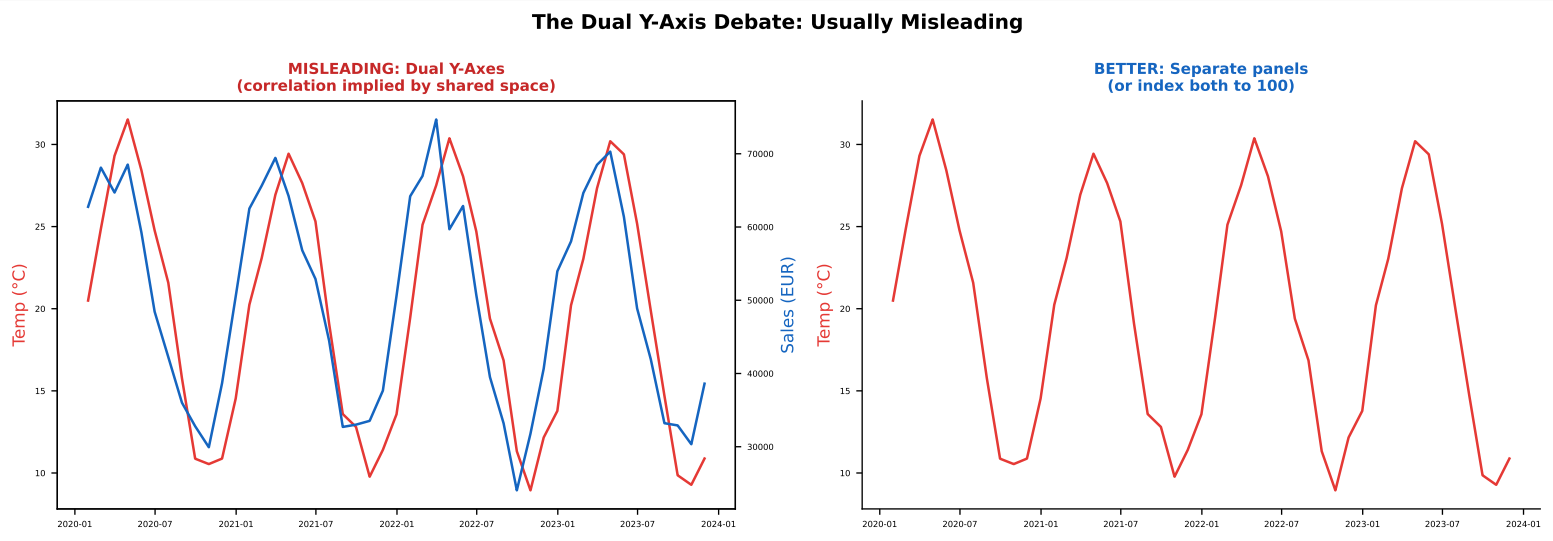


Pro-Tip

Cleveland et al. (1988) showed that slope perception is most accurate when the average slope is near 45°. Too wide exaggerates trend; too tall exaggerates noise.

Dual Y-Axes

The Dual Y-Axis Debate

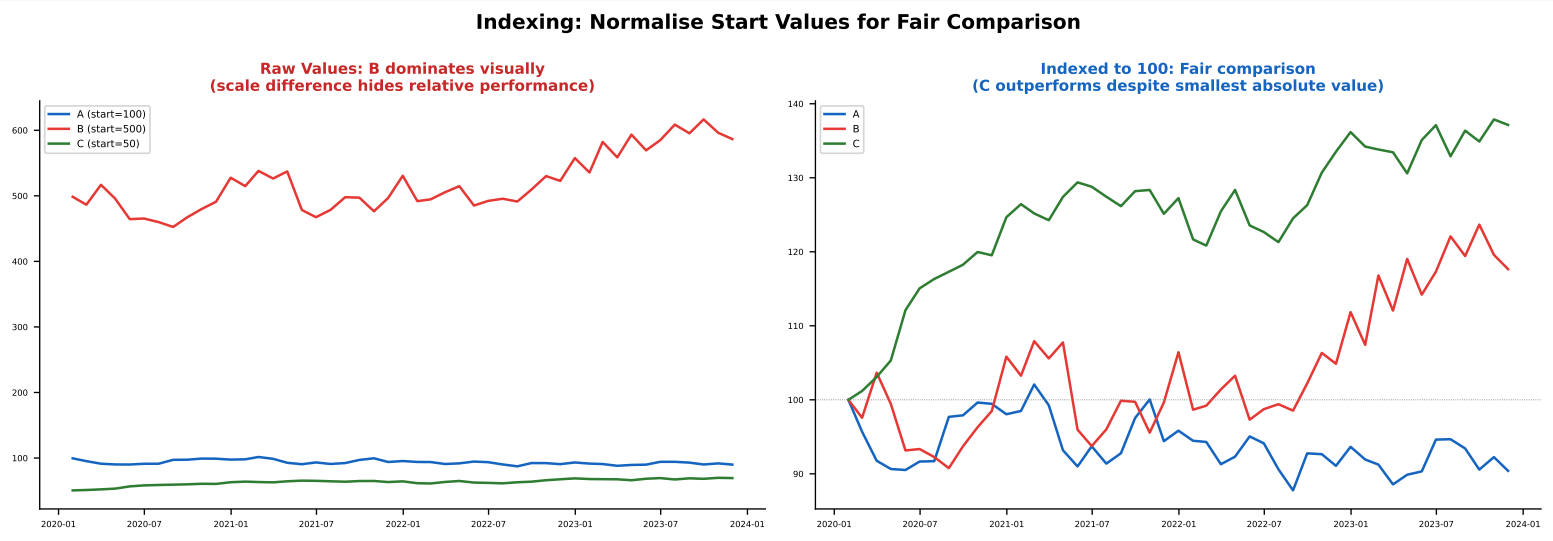


Why Dual Axes Mislead

- The **two scales are arbitrary**: changing either scale changes the apparent relationship
- Viewers **assume shared space = shared scale**, inferring correlation from visual overlap
- **Better alternatives**: (a) two separate panels with shared x-axis, (b) index both to 100, (c) secondary encoding (colour, size) instead of a second axis

Indexed Lines

Normalise Start Values for Fair Comparison

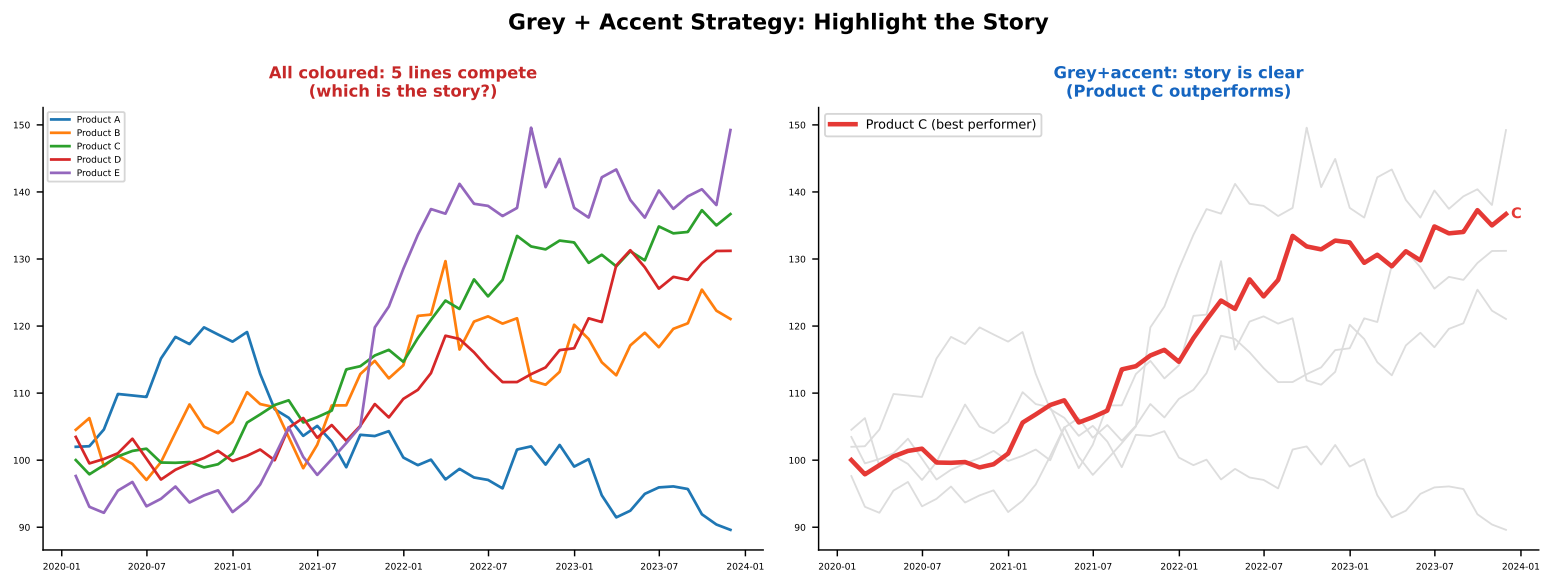


When to Index

Situation	Method	Formula
Different start values	Index to 100	$100 \times x_t / x_0$
Different units	Z-score each	$(x_t - \bar{x})/s$
Same unit, same start	Raw values fine	No transformation
Growth comparison	Log scale	Constant growth = straight line

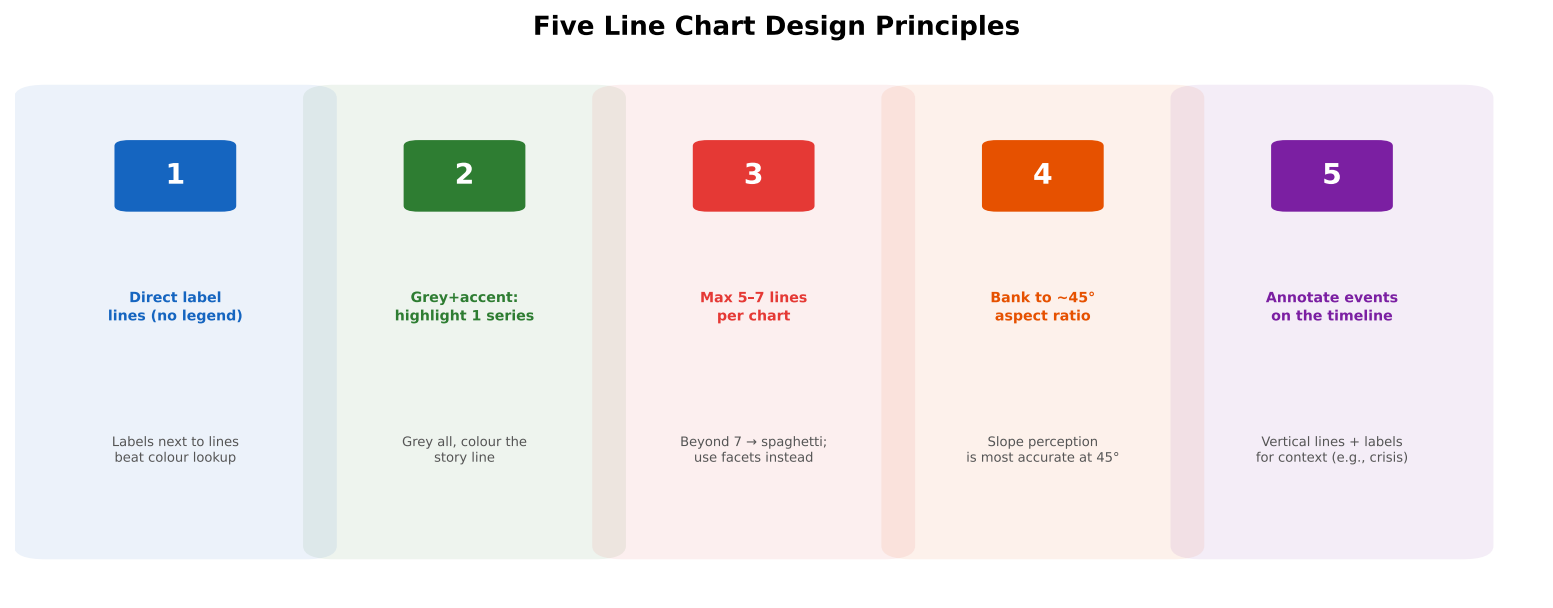
Grey + Accent

Highlight the Story



Design Principles

Five Line Chart Principles



Direct Labels and Event Annotations

R

```r
ggplot(df, aes(x=date, y=value))
    +
  geom_line(color = "#1565C0") +
  # Direct label at end
  geom_text(data = df |>
    filter(date == max(date)),
    aes(label = series),
    hjust = 0, nudge_x = 5) +
  # Event annotation
  geom_vline(
    xintercept = as.Date("
        2020-03-11"),
    linetype = "dashed",
    color = "#E53935") +
  annotate("text",
    x = as.Date("2020-04-01"),
    y = 90,
    label = "Pandemic declared",
    size = 2.5, color = "#E53935"
        )
```

Python

```python
ax.plot(dates, values, "#1565C0")

# Direct label at end
ax.text(dates[-1], values[-1],
    f" {name}", fontsize=8,
    fontweight="bold",
    color="#1565C0", va="center")

# Event annotation
ax.axvline(
    x=pd.Timestamp("2020-03-11"),
    ls="--", color="#E53935")
ax.annotate("Pandemic declared",
    xy=(pd.Timestamp("2020-03-11"
        ),
        90),
    fontsize=7, color="#E53935")
```

Synthesis

Key Takeaways

1. **Bank to** 45°: aspect ratio controls perceived slope.
2. **Avoid dual y-axes**: two arbitrary scales imply a relationship that may not exist. Use separate panels or index.
3. **Index** when comparing series with different starting values: $100 \times x_t/x_0$.
4. **Grey+accent**: all lines grey, highlight the story line.
5. **Direct label** lines at endpoints; annotate key events with vertical lines and text.

Essential Readings

- Cleveland, W. S. et al. (1988). "The Shape Parameter of a Two-Variable Graph." *JASA*, 83(402).
- Schwabish, J. (2021). *Better Data Visualizations*.
- Few, S. (2006). "Dual-Scaled Axes in Graphs."

Next Module: Line Charts in Code (W06-M03)